

Maxwell Yang

971-266-3564 | yangmax@oregonstate.edu | [linkedin.com/in/maxwellyang](https://www.linkedin.com/in/maxwellyang) | github.com/maxy1991991

EDUCATION

Oregon State University

Corvallis, OR

Bachelor of Science, Computer Science

Expected June 2027 (**3.98 GPA**; Honors College Dean's List)

- Clubs: ACM (Association for Computing Machinery), OSU International Collegiate Programming Contest Team

EXPERIENCE AND RESEARCH

SWE Intern

May 2026 – July 2026

International Business Machines (IBM)

Baton Rouge, LA

Undergraduate Research Assistant

August 2025 - May 2026

Oregon State University

Corvallis, OR

- Designed a self-evolving prediction agent that improves accuracy by 4.4% through a dynamic **RAG** based system
- Implemented **vector embeddings** to retrieve past failure cases and synthesize guidelines for future predictions
- Designed a probabilistic weight update mechanism that automatically reinforces successful reasoning patterns and decays harmful biases based on Brier score feedback
- Integrated heterogeneous **LLMs** (OpenAI/Gemini) for specialized roles in reasoning and core prediction tasks

PROJECTS

Personal Portfolio Website | *React, TypeScript, CSS, Vite*

June 2026

- Built an interactive portfolio website using **React** and **TypeScript** to showcase experience, projects, and technical skills
- Integrated an AI-powered chat bot using **Google Gemini API** to answer questions about background and experience through a knowledge system
- Implemented client-side routing with React Router for smooth page transitions and navigation between sections

Relational Database Management System | *Python, MySQL*

June 2025

- Designed a relational database schema normalized to **3NF**, enforcing referential integrity and foreign key constraints through DDL statements to eliminate redundancy
- Created single-column and composite indexes to reduce I/O overhead and improve read latency
- Managed database concurrency using explicit locks to eliminate dirty reads and maintain isolation, ensuring **ACID-compliant** transaction execution.

Transformer-based Autoregressive Language Model | *Python, PyTorch, NLP*

March 2026

- Implemented an autoregressive Transformer LM using **PyTorch**, trained to generate coherent children's stories
- Developed periodic positional encoding layers and integrated causal masking to ensure autoregressive properties during multi-head self-attention
- Engineered and compared decoding strategies, including Argmax, Temperature-scaled Sampling, and Nucleus (Top-p) Sampling, to optimize the balance between text coherence and creativity
- Evaluated model performance across varied prompts, analyzing the impact of temperature scaling and cumulative probability thresholds on distribution shift and output repetition

Shell Implementation | *Unix, C*

April 2025

- Built a custom Unix-style command-line shell in **C**, supporting foreground/background process execution, I/O redirection, signal handling, variable expansion, and built-in commands
- Managed child process cleanup and zombie prevention, tracking background jobs and their exit status
- Implemented custom signal handling for SIGINT and SIGTSTP, allowing proper process termination.

TECHNICAL SKILLS

Languages: Java, Python, C++, C, R, JavaScript, SQL, HTML/CSS, TypeScript

ML & Data Analytics: PyTorch, Autogen, Tableau, Power BI

Tools and Frameworks: React, Vite, FastAPI, Playwright, Vitest, Git, Linux/Unix, PostgreSQL

Techniques: LLMs, RAG, ML pipelines, RESTful APIs

ACTIVITIES

- * ICPC Oregon Division 2 Winner, USA Computing Olympiad Gold Division, AIME Qualifier